

INFOGRAPHIC MICRO PLASTICS



BACKGROUNDS

The prevalence of microplastics has been rapidly increasing, posing serious environmental concerns. It has been predicted that the mass of microplastics will exceed the weight of fishes in the ocean by 2051, underscoring the pressing issue that microplastics pose in the ocean. Microplastics can be easily ingested and are not biodegradable and they can bioaccumulate in marine and human life.

TYPES OF MICROPLASTICS

- Primary: Intentionally manufactured small, like micro-beads in cosmetics or plastic resin pellets (nurdles) for industry.
- Secondary: Result from the breakdown of larger plastic items (bottles, bags, tires) through sunlight, waves, and wear.

HUMAN IMPACTS

- Microplastics can be found in the food we eat, ranging from seafood to crops.
- Microplastic exposure could be linked to cancer, heart attacks and reproductive problems, but the full impacts of microplastics on the human body is still unknown.

ENVIRONMENTAL IMPACTS

- Similar to humans, microplastics also affect entire ecosystems, facilitating biofilm formation and degrading soil.
- Concentrations of microplastics can increase up trophic levels due to bioaccumulation, and can affect the health of animals

SOLUTIONS



Individual Actions

- Reduce single-use plastics
- Choose natural fibres for clothes
- Avoid nonstick and plastic cookware
- Seek out plastic-free toiletries and cosmetics

Policy & Innovation

- Plastic bans and regulations
- Improved waste management
- Implementation of novel microplastic filtration technologies
- Identifying enzyme from superworms for recycling polystyrene
- Micromotors for releasing radicals to degrade microplastics



CONCLUSION

Microplastics is a serious and pressing issue with many negative impacts. Although it is not possible to fully remove microplastics in our environment, by reducing both our exposure to microplastics and the amount of microplastics released into the environment, we can contain the problem to a manageable level.